	these data structures to solve complex problems. perform operations on **CO4:** *Understanding different searching and sorting algorithms, and their computational efficiencies.*	
	TOPIC/Description	**Instructional Hours**
Unit I	**Introduction to Data Structures:** data structures, Types of data structures, Pointers Introduction to Array & Strings, Operations on Arrays and Strings **Linked List:** Single, Double, Header, Circular Single and Double Linked List, Memory Representation of Linked List. All possible operations on linked list, **Applications of Linked List:** Polynomial Representation and manipulation	22.5
Unit II	**Stack:** Stacks terminology, Representation of Stacks in Memory, Operation on Stacks, Polish Notations, Translation of infix to postfix & prefix expression, Evaluation of Postfix Expression, **Recursion:** Problems on Recursion, **Applications of Stack:** Quick Sort and Tower of Hanoi Problem **Queue:** Memory Representation, Queue operation using array and linked list, Circular Dequeue and priority queue, Applications.	22.5
Unit III	**TREES:** Basic Terminologies, Representation of Binary Trees in Memory, Traversing of Binary tree, **Binary Search Tree:** Operation on Binary Search Tree, **Heap Tree:** Operation on Heap Tree, Heap Sort Method **Graphs:** Basics concepts of graphs, Representation of graphs, Graph traversals BFS and DFS	22.5

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Unit IV	**Searching Techniques:** Linear search and binary search, **Hashing:** Direct-address tables, Hash tables, Open addressing, Perfect Hashing, **Sorting Techniques:** Various sorting methods and their time complexity analysis: Insertion sort, Selection sort, Merge Sort, Bubble Sort.	22.5
References	**Reema Thareja, *Data Structures Using C*, Oxford University Press.** **Yashavant Kanetkar, *Data Structures Using C*, BPB Publications.** **Ellis Horowitz, Sartaj Sahni & Susan Anderson-Freed, *Fundamentals of Data Structures in C*, Universities Press.**	